Noman Bin Morshed

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Education

University of Dhaka, Bangladesh **2022–2026**

Bachelor of Science (Honours) in Nuclear Engineering **CGPA: 3.73/4.00**

Relevant Coursework: Nuclear Physics, Quantum Mechanics, Nuclear Fusion, Nuclear Instrumentation, Numerical Methods, Statistics, MATLAB Modeling, Nanoscience, Computer Programming (C++)

Selected QC Certifications: QSilver36 (QWorld), QCobalt (QWorld), QBronze – Qiskit (QWorld), QML Workshop (Fundacja Quantum AI), Fundamentals of Quantum Algorithms (IBM), Trotterization Challenge (PennyLane), LCU Challenge (PennyLane), Quantum Algorithms for Nonlinear Problems (WISER), Quantum Algorithms to Solve PDEs (WISER)

Research Manuscripts

[Under Review]

Simulated Quantum Annealing–Genetic Algorithm Framework for Nuclear Fuel Loading Optimization

- Developed a hybrid optimization framework integrating Simulated Quantum Annealing (SQA), Simulated Annealing (SA), deep-learning surrogate modeling, and Genetic Algorithms for PWR fuel loading pattern optimization.
- Formulated a 579-variable QUBO using `PyQUBO` and generated diverse constraint-satisfying seed populations through SQA/SA.
- Trained a residual CNN surrogate on 10,000 `OpenMC` simulations to predict k_{eff}, F_a, and F_q, reducing evaluation times from approximately 12 hours to under one second.

[Manuscript in Preparation]

QUBO-Based Benchmarking of Simulated Quantum and Classical Metaheuristics for PWR Fuel Loading Pattern Optimization

- Formulated reactor fuel loading optimization as a constrained QUBO and benchmarked SQA, SA, Random Search, Greedy Search, and Tabu Search across multiple reactor configurations.
- Evaluated performance using feasibility, success probability, constraint satisfaction, objective quality, and time-to-solution metrics.
- Developed CNN- and GNN-based surrogate models and integrated GA, ILS, ACO, and Cross-Entropy optimization strategies; results showed no clear simulated quantum advantage over strong classical baselines.

[Manuscript in Preparation]

Quantum-Mechanical Study of the Electronic Properties of $U_xPu_yO_z$ Compounds

- Conducted first-principles DFT simulations using Quantum ESPRESSO for uranium-based mixed oxides.
- Computed electronic and mechanical properties including total energy, band gap, DOS/PDOS,

stress, and force characteristics.

- Investigated structure–property relationships and benchmarked results against published literature.

Selected Research Projects

Quantum Galton Board Simulator

- Developed Qiskit implementations of normal, exponential, and Hadamard quantum walk distributions based on the Carney–Varcoe Universal Statistical Simulator framework.
- Designed diffusion-inspired quantum circuits using ancilla-assisted amplitude control and CSWAP-based spatial routing.
- Benchmarked performance under realistic IBM hardware noise models and improved fidelity through standard and AI-assisted transpilation workflows.

Experience

IBM Quantum – Qiskit Advocate ***Remote***

- Completed the official Qiskit Advocate Training Program.
- Earned IBM Quantum badges covering quantum algorithms, circuits, noise models, and quantum software development.
- Contributed to community-led projects involving quantum simulation and circuit optimization.

Bangladesh Atomic Energy Commission (BAEC) ***Industrial Training***

- Received hands-on training in nondestructive evaluation (NDE) and nuclear material characterization.
- Applied Eddy Current Testing (ECT), Magnetic Particle Testing (MPT), Radiographic Testing (RT), and Ultrasonic Testing (UT) for defect detection and material inspection.

Technical Skills

Programming	Python, MATLAB, C++, Git, Linux
Quantum Computing	Qiskit, PennyLane, D-Wave, OpenJij
Scientific Computing	OpenMC, Quantum ESPRESSO, TensorFlow, PyGAD
Engineering Tools	ANSYS HFSS, KiCad

Selected Academic Programs

Quantum Program (2025)	Washington Institute for STEM, Entrepreneurship and Research. Topics: Functional Quantum Linear Solvers, Quantum Algorithms for PDEs, Quantum Simulation, Lie Theory.
MTV Summer School (2025)	University of Michigan. Topics: Radiation Detection, Radiation Imaging, Scintillator Synthesis, Monte Carlo Simulation Methods.
Intro to Plasma and Fusion (2026)	Princeton Plasma Physics Laboratory (PPPL) and Oak Ridge National Laboratory (ORNL). Topics: Plasma Physics, Magnetic Fusion, Fusion Materials, AI for Fusion.